Lab Assignment 5: Filter Bank Design

Objective

1. **To understand how to construct a triangular filter**
2. **To learn how multiple triangular filters form a filter bank**
3. **To design and visualize a Mel Filter Bank**
4. **To understand the need for FFT bin mapping**

Objective1: Construction of Triangular Filters

Background: A triangular filter is a simple function defined by three points:

- Left point → where filter starts (value = a)
- Center point → peak (value = b)
- Right point → end (value = c)

i.e, From a to b → value increases linearly (rising edge)
From b to c → value decreases linearly (falling edge)
Outside → zero

Then the triangular filter $H(k)$is:

$$H(k) = \begin{cases} 0 & k < a \\ \dfrac{k-a}{b-a} & a \le k \le b \\ \dfrac{c-k}{c-b} & b \le k \le c \\ 0 & k > c \end{cases}$$

Task-1:

(i) Construct a single triangular filter with points = [(10, 20, 30)] (i.e. a=10, b=20, c=30) and plot the triangular filter.

(ii) Construct multiple triangles with points = [(10, 20, 30), (20, 30, 40), (30, 40, 50)] and plot multiple triangular filter.

(iii) Change $a, b, c$and observe shape changes. make a wider triangle and make a narrow triangle

Objective2: Design of Mel filter bank

Steps to follow:

Step 1: Define Parameters

- Sampling frequency: $f_s = 16000\ Hz$
- FFT size: $N = 512$
- Number of filters: $M = 20$

Step 2: Convert Frequency Limits (f_{min} and f_{max}) to Mel Scale ($MEL(f)_{min}$ and $MEL(f)_{max}$

Step 3: Generate Equally Spaced Mel Points ($\triangle MEL$)

Step 4: Convert Mel Points Back to Frequency ($Find\ all\ f_i$)

Step 5: Convert Frequencies to FFT Bins using the following formula

$$b_i = \frac{(N+1)f_i}{f_s}$$

Step 6: Construct Triangular Filter

For each filter $H_m(k)$:

$$H_m(k) = \begin{cases} 0 & k < b_{m-1} \\ \dfrac{k - b_{m-1}}{b_m - b_{m-1}} & b_{m-1} \leq k \leq b_m \\ \dfrac{b_{m+1} - k}{b_{m+1} - b_m} & b_m \leq k \leq b_{m+1} \\ 0 & k > b_{m+1} \end{cases}$$

<u>Report the following:</u>

(i) Plot FFT Bin vs Frequency plot (X-axis → FFT bin index, Y-axis → actual frequency)

(ii) Plot Mel Points

(iii) Print the triangular filter points (values of a, b, c for all filters)

(iv) Plot Mel filter bank

(v) Change the parameters and visualize the results

NOTES:

Why Convert Frequency to FFT Bins?

FFT output is indexed as: $X[0], X[1], X[2], \ldots, X[N-1]$

Each index corresponds to:

$$f_k = \frac{kf_s}{N}$$

f_k is the actual frequency (in Hz) represented by the k-th FFT bin. Filters must be aligned with bin indices bins during implementation.

The middle value is centered in the Mel scale, but not necessarily centered in FFT bins (or Hz).